

The experiences that places make available to people ... are an inheritance that has been entrusted to our care. ... So getting good at replenishing the places around us will just need a small stretch in our understanding.

Tony Hiss, *The Experience of Place*, 1990, p. 223

RESEARCH

1 PHENOLOGY of landscape plants in Beijing, China

Temperate Climate Plant Phenology Planting Design Preference

2 SUMMERTIME THERMAL EXPERIENCE in residential public spaces in Hong Kong

Subtropical Climate Thermal Experience Urban Heat

PRACTICE

3 SENSES HILL+VALLEY Kwun Tong Sensory Garden Renovation

Subtropical Climate Multisensory Experience Urban Renovation

4 GREEN GARDEN Greenhouse Design for 2nd Hebei Garden Expo

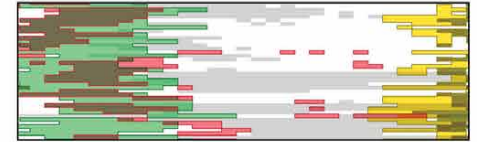
Indoor Climate Garden Design Planting Design

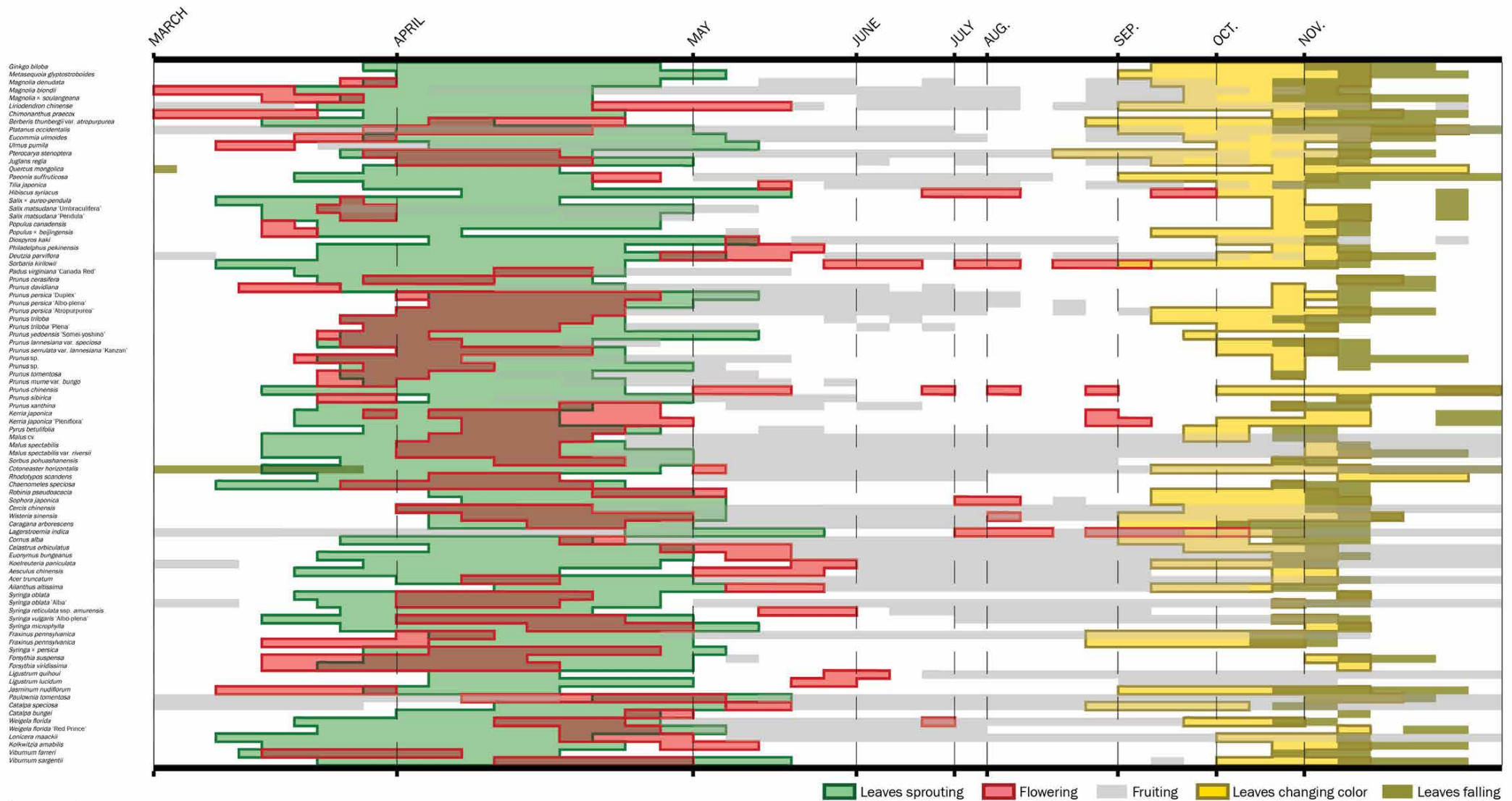
5 PLANTING DESIGN of a Rain Garden in Beijing Forestry University

Temperate Climate Rain Garden Planting Design

6 PLANTING DESIGN of quarry gardens in Nanning Garden Expo

Subtropical Climate Quarry Garden Ecological Restoration Planting Design





1 PHENOLOGY of landscape plants in Beijing, China

Seasonal change Planting design Phenophase Temperate climate

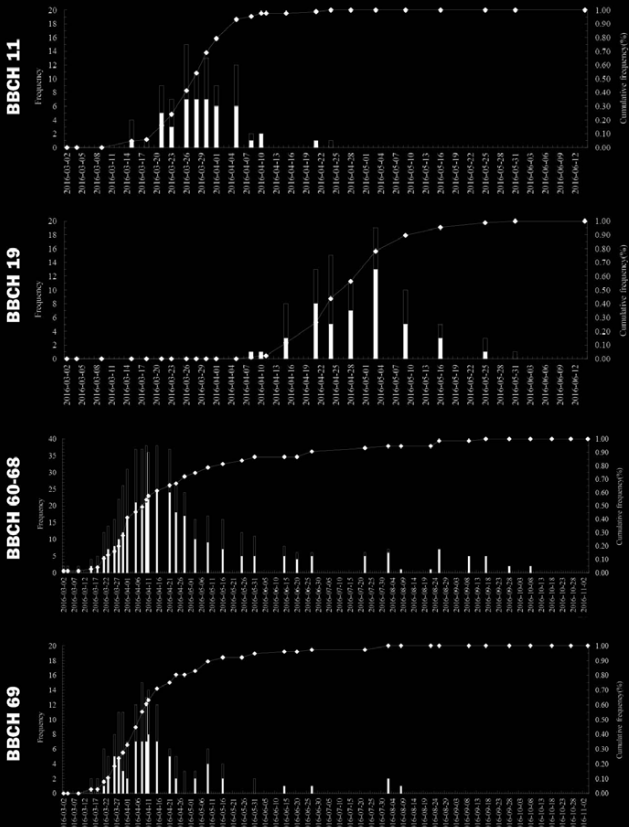
Phenological changes of vegetation bring diversity to the urban landscape, especially in temperate regions. By taking Beijing as a study case, this research project evaluated the phenological changes of landscape plants at both individual and plant community level.

Undergraduate thesis submitted to Beijing Forestry University in 2017.



PHENOLOGY TIMELINE
of observed species and cultivars

The image on the left demonstrates the phenology timeline of the 89 woody species and cultivars, based on the observations made in 2016 growing season. The different colors refer to the key ornamental states, i.e., leaves sprouting, flowering, fruiting, leaves color changing and falling. The sequences and durations of these periods are useful information for selection of plant species in planting design.



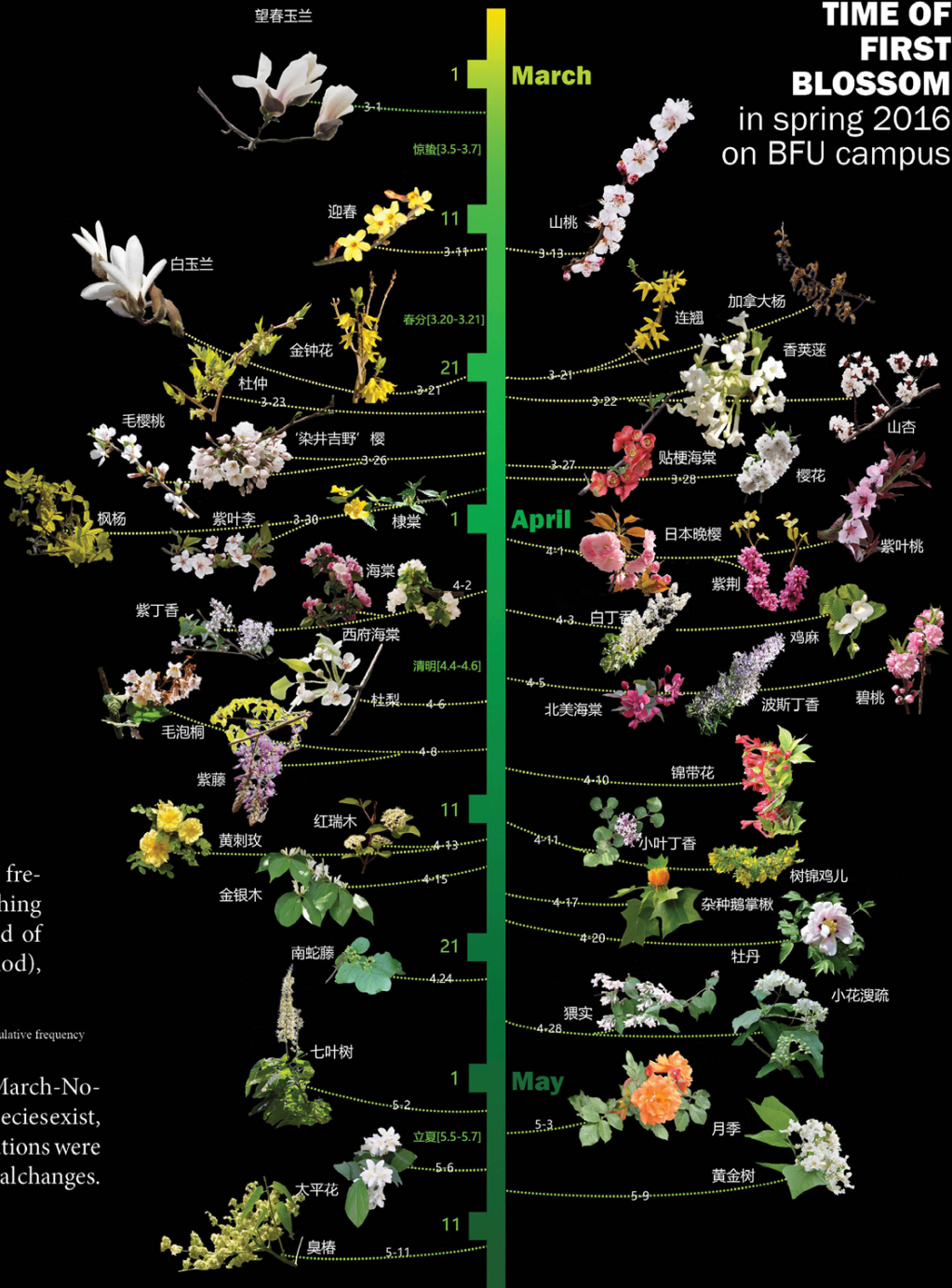
TIME OF KEY
PHENOPHASES:
leaves emergence and
blossom

The charts illustrate the accumulative frequencies of species and cultivars reaching BBCH 11 (emergence of leaves), 19 (end of leaves emerging), 60-68 (flowering period), and 69 (end of blossom).

Tree Shrub/Vine Cumulative frequency

Phenology observations of 89 widely-used woody plants were conducted during growing season in 2016 (March-November). BBCH scale was applied to record the phenological stages. On the campus of BFU, a variety of plant species exist, making it a good phenology garden for phenological observations. During spring and autumn, the observations were made at an interval of 3-5 days. While in summer, longer interval of 7-9 days was adopted due to fewer phenological changes.

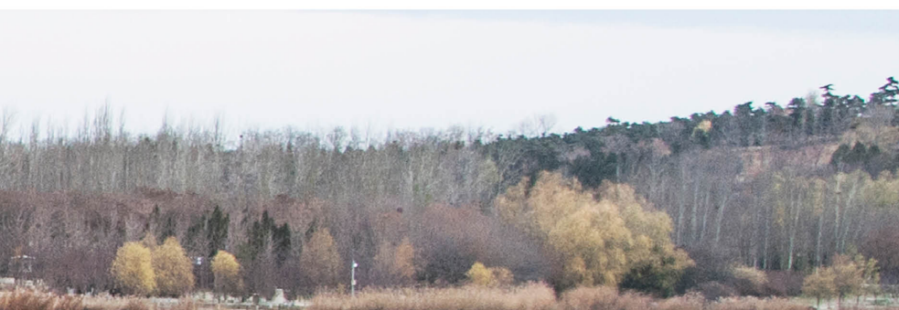
PHENOLOGICAL OBSERVATIONS
at individual level of 89 commonly used
woody species and cultivars





HUMAN PERCEPTION ON SEASONALITY of plant community

In practice, plants are rarely planted individually. Rather, they form plant communities that demonstrate seasonality across different seasons. This section evaluates the perceptions of seasonality of landscape and how the plant characteristics may influence such human perception.



Overview

This study investigates the seasonal dynamics of urban park plant communities in Beijing. Using Scenic Beauty Estimation (SBE) method, we evaluated the aesthetic values of 25 representative plant communities across four seasons. By unpacking various community characteristics, we established multiple linear regression models to identify the key driving factors behind public aesthetic preferences across all four seasons, offering quantitative scientific evidence for planting design.

Data

[Image Collection] Seasonal landscape photographs of 25 plant communities with healthy growth and high-quality maintenance across 5 urban parks.

[Subjective Evaluation (SBE)] Rated with slide projections with 60 third-year landscape architecture students. Evaluators used a 7-point scale ranging from -3 (Very Poor) to +3 (Excellent).

[Plant Community Characteristics] 15 plant community composition factors (e.g., Simpson diversity indices, evergreen ratios, shrub ratios, phenological ratios) and 5 structural/aesthetic appearance factors (e.g., growth status, space creation, number of plant layers, color counts).



Result

Aesthetic appreciation is highly dependent on distinct structural and visual variables across different seasons. Shrub layer characteristics and the total number of plant layers consistently exert the highest influence on overall scenic beauty.

	SPECIES COMPOSITION						STRUCTURE		
SPRING	+					+	-		+
SUMMER						+			+
AUTUMN		+							+
WINTER		+							+
	Simpson's Index			Evergreen %	Cone %	Shrub %	Flowering / coloration tree %	Flowering / coloration shrub %	No. plant layers
	Tree	Shrub	Herb						Color diversity
									Plant growth status

Takeaway

[Enhance Summer Color Contrast] Break summer monotony by contrasting light-green foliage (*Catalpa speciosa*, *Kerria japonica*) against dark-green backdrops (*Pinus tabulaeformis*, *Sabina chinensis*), supplemented by colorful foliage or summer blooms.

[Optimize the Shrub Layer] Focus on close-range human scales by arranging highly diverse shrub communities that display rich seasonal fruit, leaf transitions, and unique branching forms.

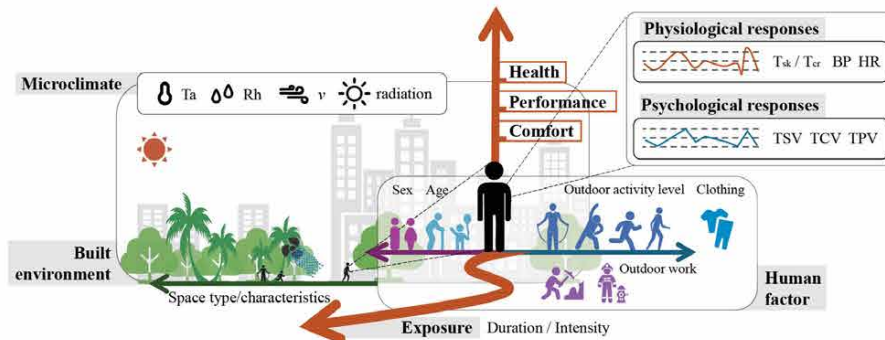
2 SUMMERTIME THERMAL EXPERIENCE in residential public spaces in Hong Kong

Outdoor thermal comfort Thermal alliesthesia Transient thermal experience Heat stress Climate responsive design

Context

Rising extreme heat events pose challenges to designing healthy and comfortable outdoor public spaces of high environmental quality, especially in subtropical high-density urban contexts. The diversity of outdoor thermal environment offers a potential solution to future-heat-resilient urban and landscape design, particularly when considering human adaptations and dynamic behaviors in outdoor environments.

However, we are not fully aware of the status quo of outdoor thermal environment diversity under current extreme heat, or the pathways influencing human physiological and psychological experiences. An evaluation workflow to assess the dynamic outdoor thermal experiences under current or future scenarios is also missing. Such evidence and tools are necessary to support climate-resilient evidence-based design, which requires context and climate-zone specific suggestions and guidelines.



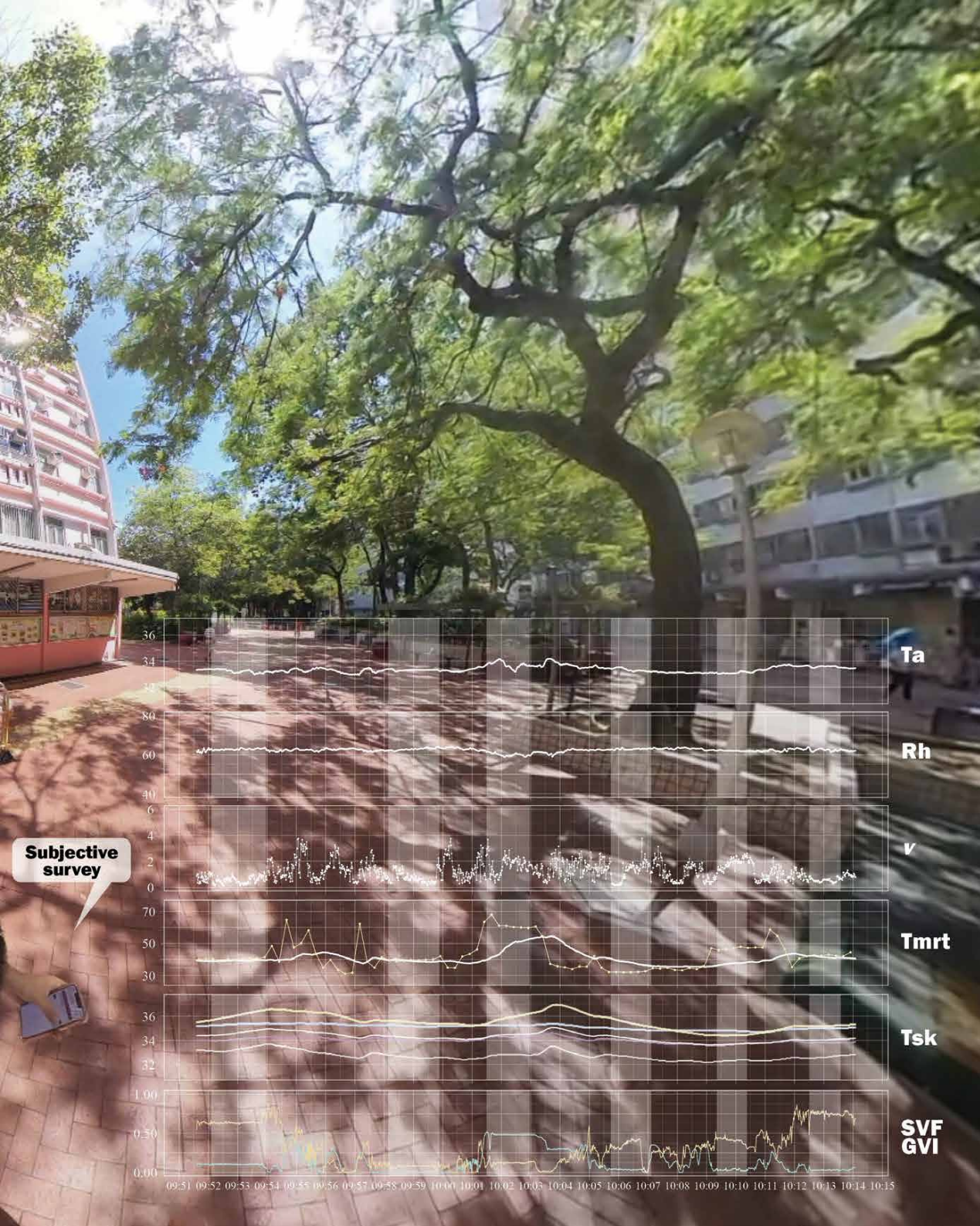
This thesis addresses these gaps by taking public housing estates (PHEs) in subtropical high density Hong Kong as case studies. Within this urban context, thermal environments and individuals' thermal experiences in three typological types of public spaces, i.e., open squares, vegetated spaces, and semi-outdoor spaces, were evaluated, with **(1)** microclimate measurements conducted to characterize the diversity of outdoor thermal environments under current extreme heat, **(2)** walking-based surveys of thermal and environmental perceptions conducted simultaneously with built environment, microclimate variables and human skin temperature measurements to build the influencing pathways, **(3)** future extreme heat scenarios simulated with ENVI-met and human outdoor thermal experiences modeled with JOS-3, and **(4)** context-specific guidelines developed based on scientific evidence.

PhD thesis submitted to The University of Hong Kong in 2025.



Lai Kok Estate
Cheung Sha Wan

Mobile
measurement



Site: Public Housing in Hong Kong

Hong Kong's PHEs are selected as the cases to evaluate the dynamic thermal experiences under the current summer heat and answer the research question context specifically. After decades of development, Public Housing Estates in Hong Kong now accommodates over 40% of Hong Kong's population, making them an integral living environment.

Throughout this PhD project, eight PHEs were selected as study sites, considering their construction year, location, typical building typology, and abundance of different types of public spaces. Constructed between 1960s - 1990s, these PHEs represent the old and middle-aged Public Housing Estates in Hong Kong, which will be prioritized in urban renewals.

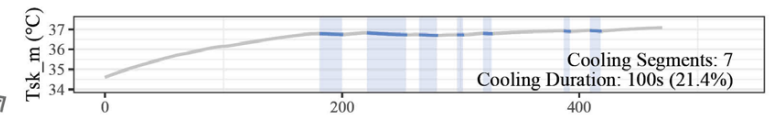
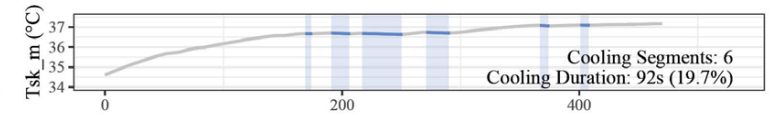
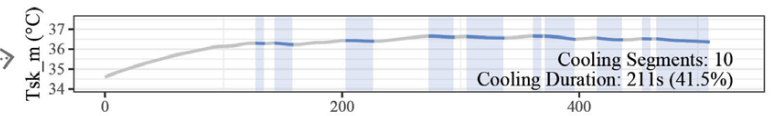
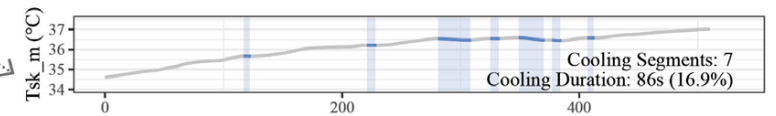
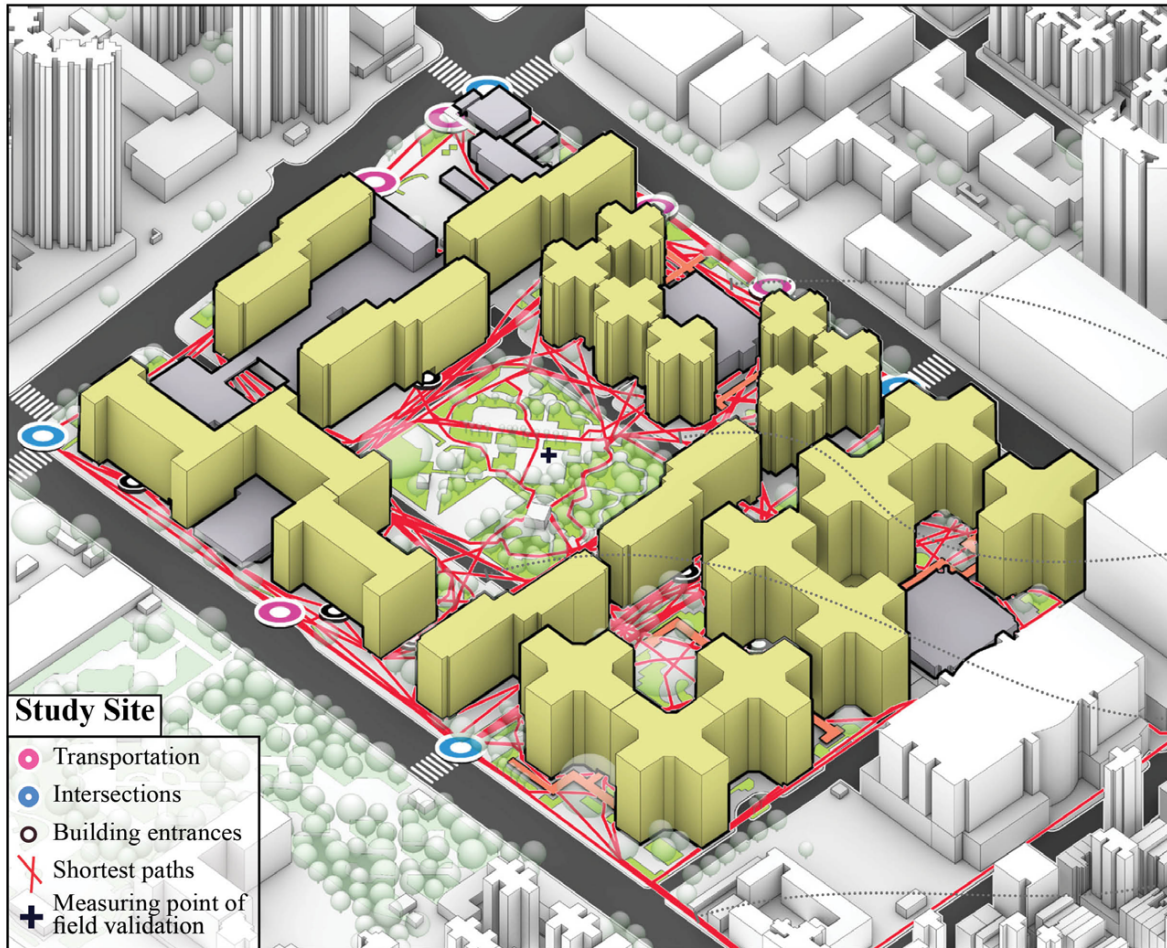


OPEN
SPACES

VEGETATED
SPACES

SEMI-
OUTDOOR
SPACES

Shaped by the buildings in PHEs, open squares, vegetated spaces, and semi-outdoor spaces are typical types of public spaces in PHEs, featuring distinct functions, usage patterns, morphological characteristics. The generic design of building types and outdoor public spaces have resulted in relatively higher satisfaction among the residents living in Hong Kong's PHEs. These typical types well summarize the public space typologies and provide key social benefits to the residents, making knowledge generated from these case studies generally transferable.



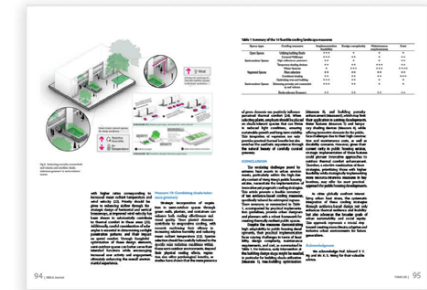
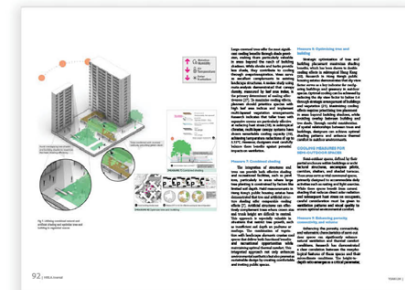
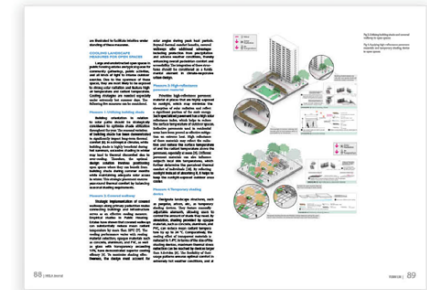
Skin temperature profile along selected paths at 13:00 in 2040 under SSP5 scenario. Variations in skin temperature along paths demonstrates evidence of thermal alliesthesia.

Thermal experience under future extreme heat

While transient thermal comfort in outdoor environments is widely documented, tools for assessing dynamic outdoor thermal experiences are needed to assist the design of future-heat-resilient public spaces.

This study demonstrates an integrated assessment workflow combining ENVI-met microclimate modeling, shortest path generation, and JOS-3 human physiological response modeling, using Hong Kong's public housing estates as a study case.

Modeling of human thermal experiences in 2040 extreme heat revealed moderate yet safe heat stress; variations in skin temperature changes along walking routes demonstrated evidence of thermal alliesthesia, with higher frequency along longer walking routes. The evaluation was conducted using Grasshopper, demonstrating its practical application as a tool for urban and landscape design.



Practical Suggestion

Based on these findings and an umbrella review, ten feasible cooling measures were illustrated for typological public spaces in Hong Kong's PHEs, accounting for the subtropical climate, high-density urban context, cost effectiveness, and local applicability. These measures focus on key microclimate parameters that influence thermal comfort, e.g., reducing solar radiation, enhancing pedestrian level ventilation, etc. We advocate for the integration of these cooling strategies in future public space design to ensure both spatial diversity and improved thermal comfort, enabling residents to better adapt to increasing future extreme heat conditions.

Conclusion

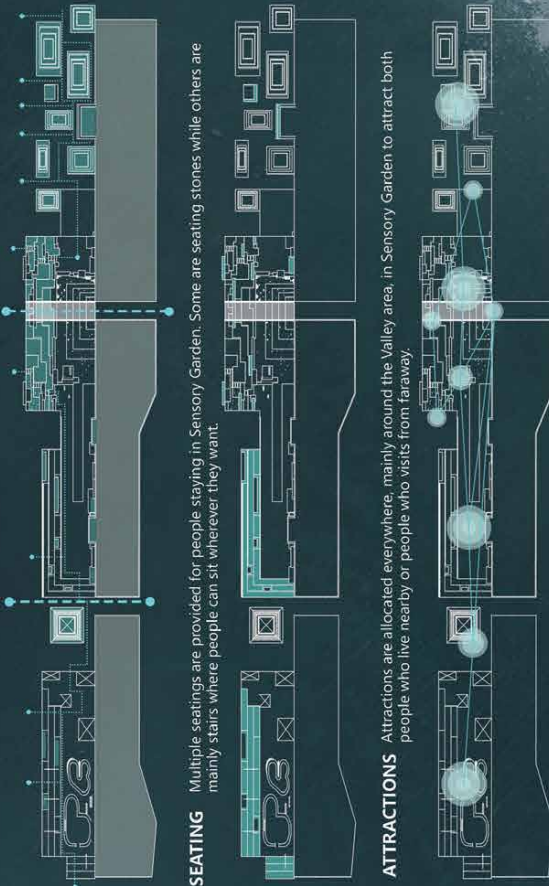
This thesis emphasizes the significance of diversity in outdoor thermal environments and experiences under extreme heat. Such diversity can be harnessed through evidence-based design and assessed using modeling approaches, providing pathways for designing climate-resilient public spaces in the face of future extreme heat.

3 SENSES HILL+VALLEY 感觀丘谷

Urbanized Natural Experience Scheme
Renovation of Kwun Tong Sensory Garden
城市自然體驗計劃
觀塘感官花園設計提升方案

Three elements in nature which are closely related to human perception were extracted, namely "cloud", "wind" and "mist". It exhibits nature experience in a highly urbanized coastal context through the means of abstraction and space creation, with human sensory integrated with different landscape elements. This conceptual design proposal provides 7/24 multi-functional spaces for the Kwun Tong community.

Merit Prize of student group, 2022.



DESIGN ANALYSIS



0 5 15 25 m

- 1 Garden of CLOUD
雲丘花園
- 2 Gate of MIST
霧谷之門
- 3 Garden of MIST
霧谷花園
- 4 MIST hang bridge
霧谷懸橋
- 5 Community Plaza
觀塘聚場
- 6 WIND Energetic
活力風丘



SITE LOCATION

Sub-tropical Northern Hemisphere
Hong Kong SAR
China

Kowloon Peninsula
Sea-side
Hong Kong SAR

Kwun Tong

HISTORY

Plastic products were massively produced in Kwun Tong area

Plastic flowers production

Textile industry once flourished

Textile industry once flourished

SITE CONDITION

LAND USES

KWUN TONG ROAD

500m

250m

KWUN TONG BYPASS

KWUN TONG PROMENADE

SITE

Legend:

- Residential Buildings (Yellow)
- Industrial Buildings (Purple)

Landmarks: Longhua Park, Kwun Tong Station, Kwun Tong Ferry Pier, Kwun Tong Market, Kwun Tong Police Station, Kwun Tong Fire Station, Kwun Tong Hospital, Kwun Tong Cemetery, Kwun Tong Sports Ground, Kwun Tong Swimming Pool, Kwun Tong Tennis Centre, Kwun Tong Basketball Court, Kwun Tong Football Field, Kwun Tong Hockey Field, Kwun Tong Badminton Court, Kwun Tong Table Tennis Centre, Kwun Tong Judo Hall, Kwun Tong Karate Hall, Kwun Tong Taekwondo Hall, Kwun Tong Judo Hall, Kwun Tong Karate Hall, Kwun Tong Taekwondo Hall.

Industrial Buildings

SURROUNDING GREENERY

ROADSIDE GREENERY

SITE

KWUN TONG PROMENADE

The site is surrounded by **dense greenery** on both sides. Kwun Tong Promenade provides sufficient spaces for public. Therefore, creating attractions is crucial.

The diagram illustrates the proposed Kwun Tong Bypass route. It shows a series of green rectangular blocks representing industrial buildings. A purple line with arrows indicates the route, starting from the top right, passing through a 'PEDESTRIAN' area, and then through a 'SITE' area. The route is labeled 'KWUN TONG BYPASS' and 'KWUN TONG PROMENADE'. A label 'SURROUNDING TRAFFIC' is at the top right. A note at the bottom right states: 'Two lanes through the site connects the road to the space beneath Kwun Tong bypass. It is needed to create interesting spots along the route beneath the bypass.'

CURRENT FACILITY

Roadside Greenery

SITE!

Grey Spaces

Lacks greenery, facilities and spaces for people hanging out.

Entertainment facilities

Current facilities are out of use and not SENSE related

CURRENT FACILITY

Roadside Greenery

SITE!

Grey Spaces

Lacks greenery, facilities and spaces for people hanging out.

Entertainment facilities

Current facilities are out of use and not SENSE related

WIND Energetic
活力風丘

Hanging structure which could swing in the wind. Made from recycled plastic.

Garden of MIST
霧谷花園

Man-made mist for close interaction with people.

Garden of CLOUD
雲丘花園

Hanging structure which is an abstraction of cloud.

Garden of CLOUD
雲丘花園

C-C Section
Community Plaza
觀塘聚場

Diagram illustrating the C-C Section of the Community Plaza (觀塘聚場) design, showing the relationship between the Road Greenery, Sunken Plaza, and Terrace areas.

Key features and dimensions shown in the section:

- Fixed to Corten steel structure:** +6.40
- Lights implanted:** +4.20, +3.50, +1.10
- Hanging element of CLOUD:** +4.80, +3.90
- Made from recycled plastic:** +0.85
- Screen fixed to steel structure:** +5.10
- Stairs where people can sit:** +2.10, +1.00, +0.30
- Portable seats which can be moved away:** +0.30
- Ground Level:** ±0.00
- Dimensions:** 100m Path, 400m Hill, 120m Path, 450m Hill, 300m Terrace, 5.10m Sunken Plaza.
- Areas:** Road Greenery, Sunken Plaza, Terrace, Beneath Bypass.

Garden of MIST
霧谷花園

D-D Section

WIND Energetic
活力風丘

Mist generating structure.

Glass bridge with LED
installed which can be controlled by cell phone app.

Multiple seating spaces.

Roller skating court for the young.

Hanging structure which could swing in the wind. Made from recycled plastic.

Road Greenery Terrace Water Feature Terrace Garden Beneath Bypass

Road Greenery Skate Lot 輪滑場 Beneath Bypass

Garden of MIST
霧谷花園

D-D Section

WIND Energetic
活力風丘

Mist generating structure.

+4.40

+5.10

+2.75

+0.00

1.20m

-0.60

7.40m

Glass bridge with LED installed which can be controlled by cell phone app.

Multiple seating spaces.

Road Greenery

Terrace Water Feature

Terrace Garden

Beneath Bypass

+4.00

+3.20

+0.00

5.60m

-0.60

4.90m

Roller skating court for the young.

Hanging structure which could swing in the wind. Made from recycled plastic.

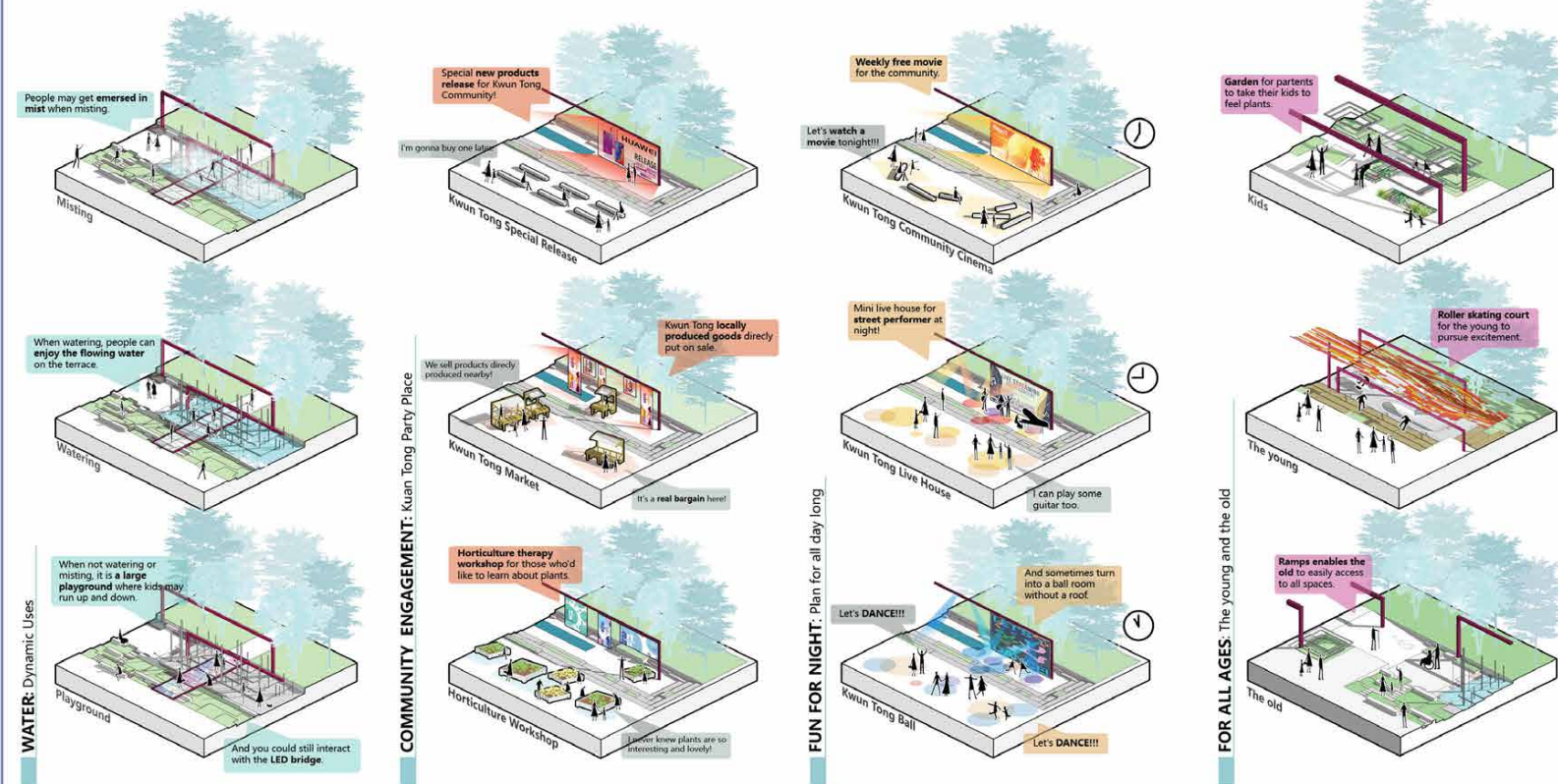
Road Greenery

Skate Lot

輪滑場

Beneath Bypass

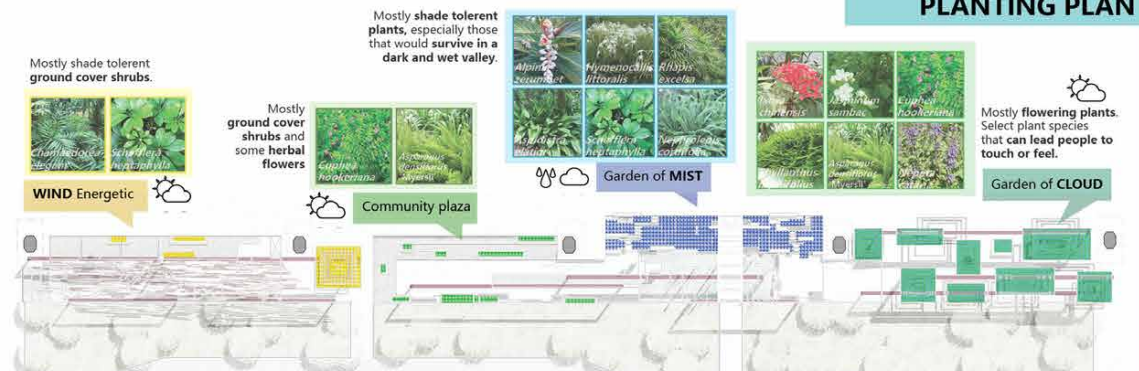
MULTI-FUNCTION SPACES



FUN KWUN TONG

	JAN.	FEB.	MAR.	APR.	MAY	JUN.	JUL.	AUG.	SEP.	OCT.	NOV.	DEC.
Garden of CLOUD												
<i>Ixora chinensis</i>												
<i>Asplenium sambac</i>												
<i>Cuphea hockieniana</i>												
<i>Phyllanthus myrsinites</i>												
<i>Nepeta cataria</i>												
<i>Alpinia zerumbet</i>												
<i>Alpinia zerumbet</i>												
<i>Rhapis excelsa</i>												
<i>Schefflera heptaphylla</i>												
<i>Nephrolepis cordifolia</i>												
<i>Commelina purpurea</i>												
<i>Chamaedorea elegans</i>												
Garden of MIST												
<i>Hymenocallis littoralis</i>												
<i>Alpinia zerumbet</i>												
<i>Rhapis excelsa</i>												
<i>Schefflera heptaphylla</i>												
<i>Nephrolepis cordifolia</i>												
<i>Commelina purpurea</i>												
<i>Chamaedorea elegans</i>												
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<i>Rhapis excelsa</i>												
<i>Schefflera heptaphylla</i>												
<i>Nephrolepis cordifolia</i>												
<i>Commelina purpurea</i>												
<i>Chamaedorea elegans</i>												

Selection of plant species are based on the different HABITAT in the HILLS and the VALLEY.







#Garden of Musa

4 GREEN GARDEN 2nd Hebei Garden Expo

With Prof. Li Dong's Studio, BFU.

Contribution: Schematic Design, Development Design of Garden of Musa

This commissioned project, completed in 2018, is located in Hebei, China. It consists of three garden spaces within a greenhouse building. Situated in northern China, the client wished to showcase plants that typically cannot survive outdoors in the temperate climate.

We therefore selected three types of plants with distinctive symbolic significance in Chinese plant culture, i.e., Musa, Bamboo, and Palm, and created three distinctly styled plant spaces. In these plant-featured spaces, combined with other landscape elements, we created cultural imagery of "rain dropping on banana leaves" ("雨打芭蕉"), "picture scroll of palms" ("棕榈长卷"), and "bamboo rustling in the wind" ("百竹萧萧").

Concept

This project orchestrates a living geographical journey across China, bringing together three iconic plant types—Musa, Bamboo, and Palm—each serving as a distinct cultural and regional symbol. This lush southern interior stands in direct dialogue with a native, open-air central courtyard showcasing northern vegetation, creating a poetic design contrast between local landscapes and the imported southern oases.

Garden of Musa

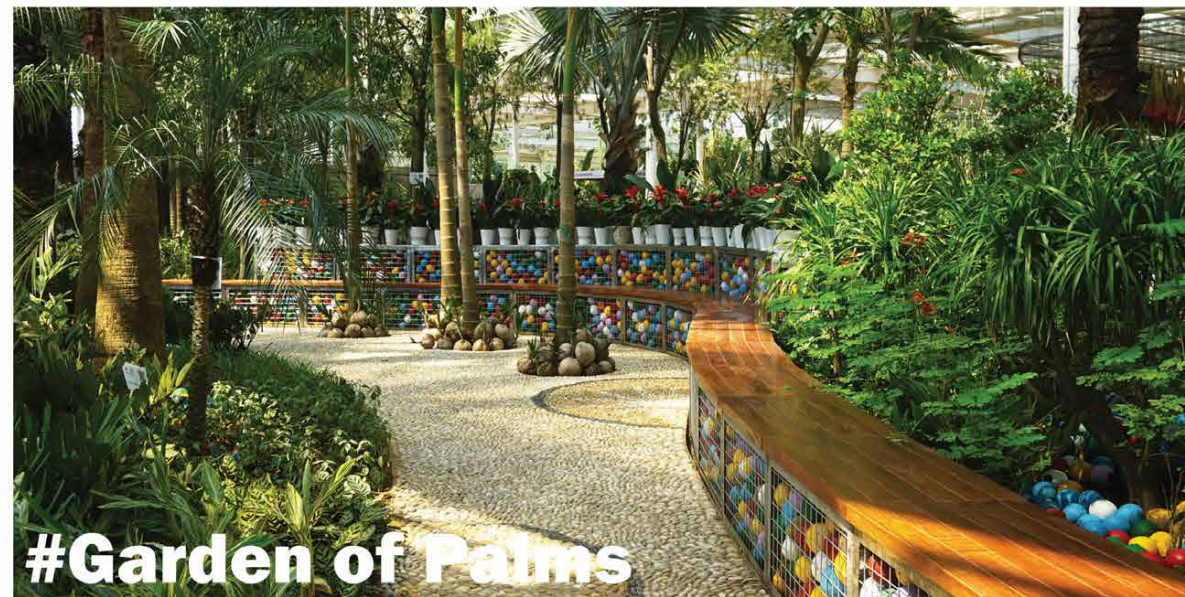
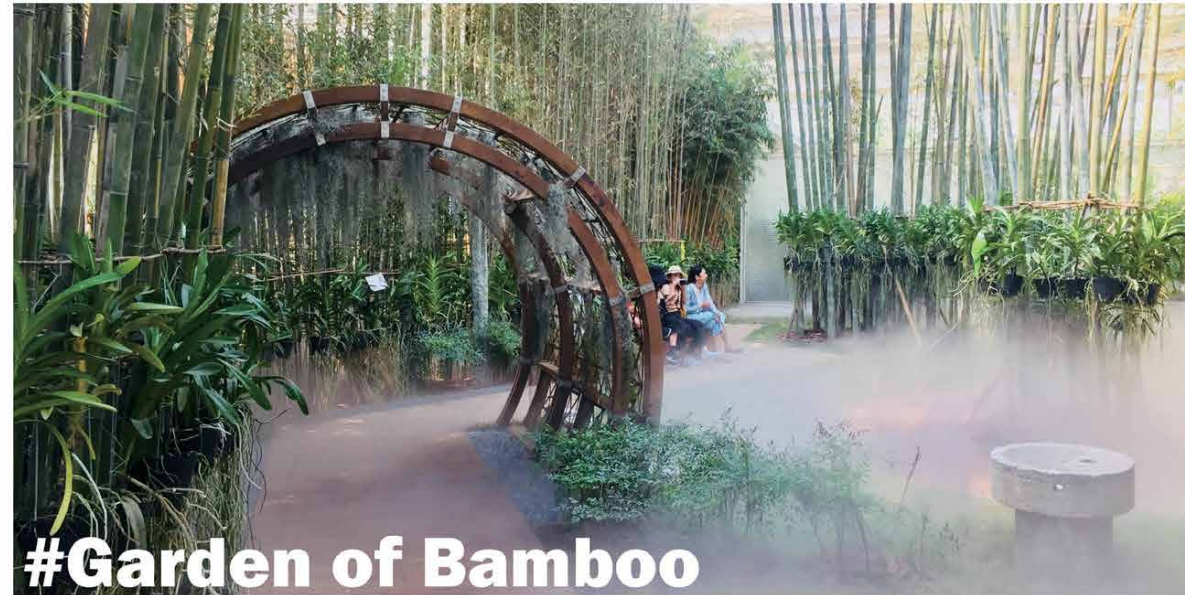
Replicating the humid water-garden typologies of the Jiangnan region, this zone features an overstory skeleton of traveler's palms, giant bird of paradise, and high-mountain figs transitioning to a understory of *Philodendron*, *Monstera*, and *Calathea*. The core layout integrates *Musa basjoo* with open water elements, traditional clay roofing tiles, and an automated overhead misting system. An abstracted, structural glass moon gate replaces traditional masonry partitions, framing the simulated precipitation to replicate the classical auditory and visual landscape effect of rain drops hitting broad musa leaves.

Garden of Bamboo

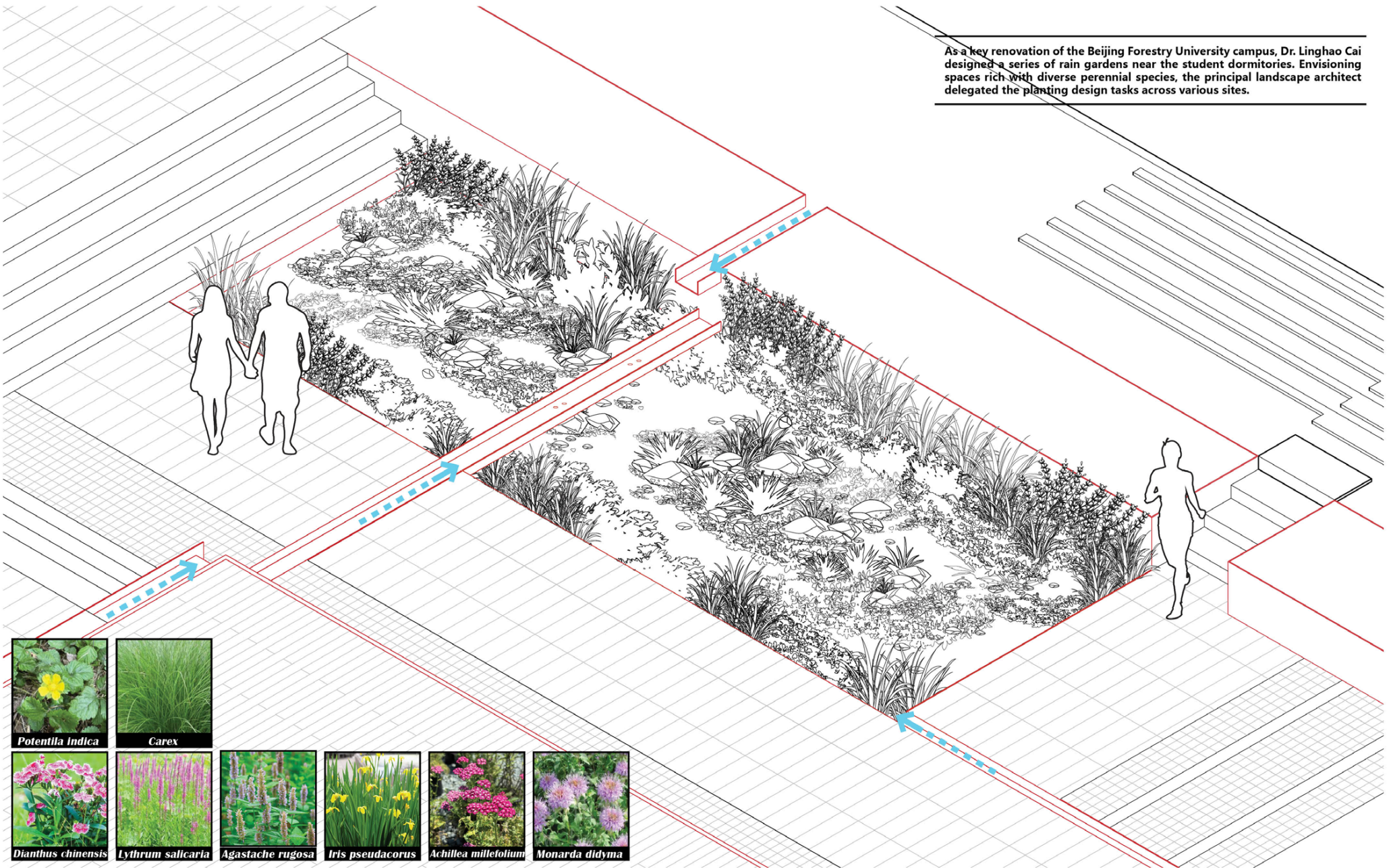
Modeling the dense groves of the Western Sichuan region, this zone utilizes a specialized matrix of running and clumping bamboo species underplanted with *Nandina domestica*. Winding timber boardwalks navigate the space, compressing the spatial layout to vary visitor sightlines through changing canopy densities and an integrated high-pressure fog system. Modern glass-brick screen walls substitute traditional courtyard partitions, leveraging material translucency to capture the back-lit shadows of the stalks and reproduce classic screening patterns.

Garden of Palms

This zone models the coastal plant communities of the southern Lingnan region through vertical canopy stratification and open visual fields. The overstory layers mature palms (*Washingtonia*, *Roystonea*, *Archontophoenix*, *Phoenix*) with prized southern hardwoods like agarwood, rosewood, and red sandalwood. Shrub layers of *Rhapis excelsa* and *Cycas revoluta* drop down to ground covers and suspended epiphytic orchids.



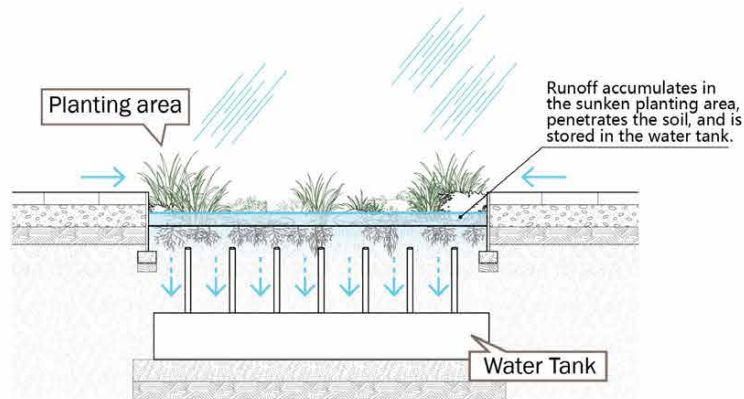
As a key renovation of the Beijing Forestry University campus, Dr. Linghao Cai designed a series of rain gardens near the student dormitories. Envisioning spaces rich with diverse perennial species, the principal landscape architect delegated the planting design tasks across various sites.



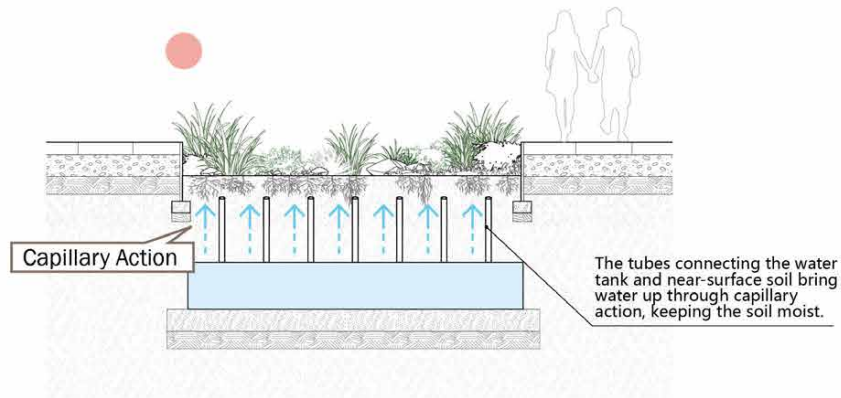
5 PLANTING DESIGN of a Rain Garden in Beijing Forestry University

With Prof. Li Dong's Studio, BFU.
Contribution: Development Design

Storing



Using



Set at a lower elevation than the surrounding pavement, this planting area actively collects surface runoff. To visually create a sense of stream flow, the principal landscape architect required intergrating crushed gravel and cobbles into the design. Functionally, the system stores excess water in an underground tank during rain, while vertical tubes use capillary action to pull moisture back up to the roots during dry spells.

This dual-phase system creates a highly variable soil environment: completely submerged during rainfall, wet immediately after, and mostly moist but occasionally dry during droughts. Therefore, the system requires resilient vegetation that can tolerate both severe waterlogging and dry soil. Based on these strict ecological requirements, species such as *Iris*, *Potentilla*, and *Achillea* were selected, ensuring the planter features continuous blossoms from late spring to autumn.

By combining this passive water-recycling infrastructure with highly adaptable plants, the design achieves both ecological sustainability and low-maintenance beauty. The capillary action drastically minimizes the need for manual surface irrigation, conserving water while ensuring the resilient plant palette thrives across changing weather conditions. Furthermore, the continuous blooming period not only enhances the pedestrian experience along the adjacent walkway but also supports local urban ecology, proving that green infrastructure can seamlessly blend advanced hydrology with striking landscape architecture.



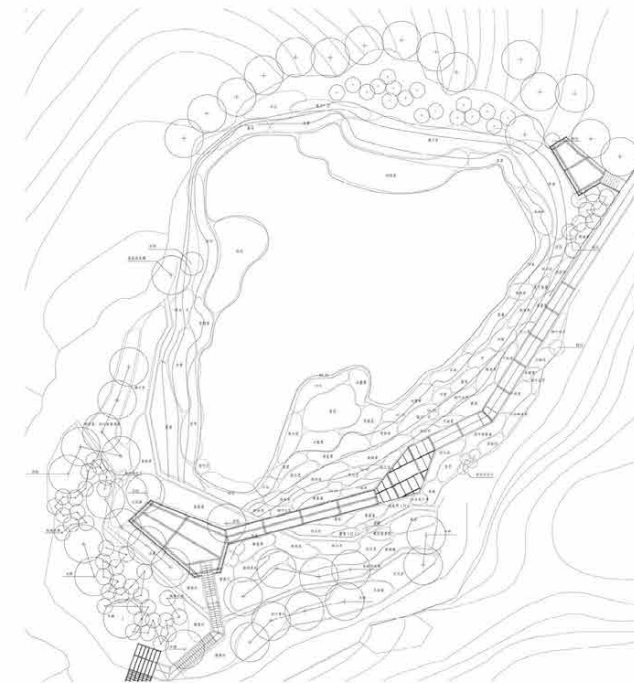


Water Garden, demonstrated here, is a compact 0.4-hectare quarry garden deliberately designed to guide visitors down to the water edge, where depths reach over 30 meters. The principal landscape architect, Prof. Xiangrong Wang, envisioned the project through a lens of minimal intervention, preserving the steep, dramatic cliffs as the most striking landscape element.

In strict alignment with the design philosophy, the planting design adopts a deliberately rugged and wild approach.

Zero intervention was applied to the cliff areas, intentionally retaining the spontaneous native species as it was to maintain its raw character. Along the elevated paths, amphibious aquatic plants capable of adapting to fluctuating water levels across both dry and flood seasons were strategically selected, including *Oenanthe*, *Scirpus*, *Lythrum*, etc.

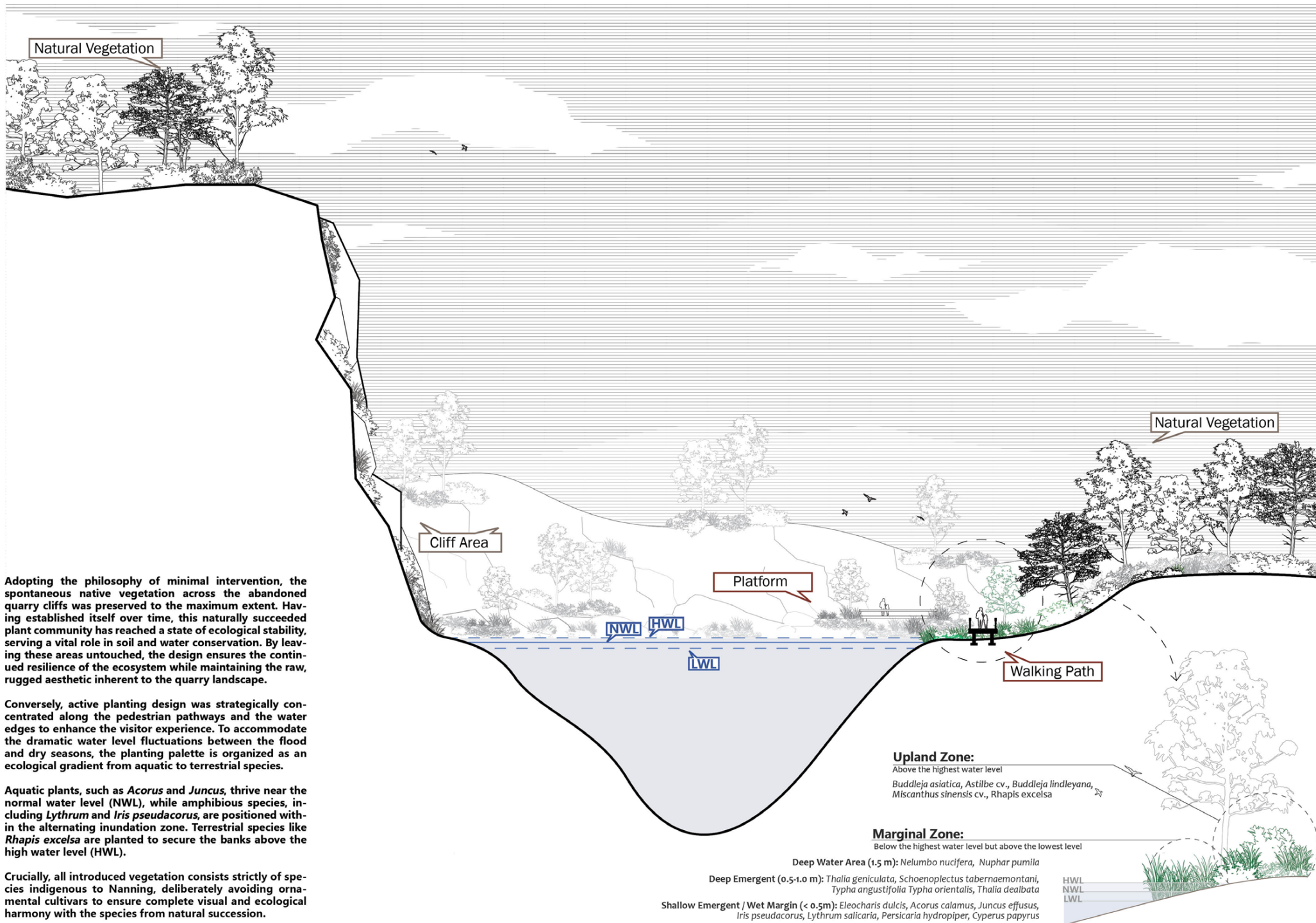
In the terrace areas, which are safely positioned above potential flood zones, a striking foreign blossom tree—the Silk Floss Tree (*Ceiba speciosa*)—was chosen as the primary focal species of this rugged quarry. Its vibrant presence serves as a deliberate design cue that anchors and highlights the beautifully untamed, messy ecosystem.



6 PLANTING DESIGN of quarry gardens in Nanning Garden Expo

With Prof. Li Dong's Studio, BFU.
Contribution: Development Design

Designed by ATELIER DYJG, the 2018 International Garden Expo in Nanning transformed a series of abandoned quarries into restored green spaces. Three quarry gardens which were planned as Water Garden, Rock Garden, and Terrace Garden were delegated for planting design.



Adopting the philosophy of minimal intervention, the spontaneous native vegetation across the abandoned quarry cliffs was preserved to the maximum extent. Having established itself over time, this naturally succeeded plant community has reached a state of ecological stability, serving a vital role in soil and water conservation. By leaving these areas untouched, the design ensures the continued resilience of the ecosystem while maintaining the raw, rugged aesthetic inherent to the quarry landscape.

Conversely, active planting design was strategically concentrated along the pedestrian pathways and the water edges to enhance the visitor experience. To accommodate the dramatic water level fluctuations between the flood and dry seasons, the planting palette is organized as an ecological gradient from aquatic to terrestrial species.

Aquatic plants, such as *Acorus* and *Juncus*, thrive near the normal water level (NWL), while amphibious species, including *Lythrum* and *Iris pseudacorus*, are positioned within the alternating inundation zone. Terrestrial species like *Rhapis excelsa* are planted to secure the banks above the high water level (HWL).

Crucially, all introduced vegetation consists strictly of species indigenous to Nanning, deliberately avoiding ornamental cultivars to ensure complete visual and ecological harmony with the species from natural succession.



with **AECOM**
Beijing office
2020-2021



Imagine it. Delivered.



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Collected in Beijing



Metasequoia glyptostroboides
水杉



Galinsoga parviflora
牛膝菊



Persicaria sp.
蓼屬

Collected in Hong Kong



Viola diffusa
七星蓮



Hypericum japonicum
地耳草



Polygala hongkongensis
香港遠志



Acacia confusa
台灣相思



Ilex asprella
秤星樹



Yilun Li